
Handwritten Digit Recognition

Using Convolutional Neural Networks (CNN)

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1. Introduction

Handwritten digit recognition is a classic image classification problem in the field of Computer Vision and Deep Learning. It involves training a model to correctly identify hand-drawn digits (0-9) from pixel data. This problem serves as an excellent entry point into Convolutional Neural Networks (CNNs), which are specifically designed to exploit the spatial structure of images.

The MNIST dataset (Modified National Institute of Standards and Technology) is the standard benchmark dataset for this problem, containing 70,000 grayscale images of handwritten digits at 28x28 pixel resolution. It has been widely used in research and education to validate machine learning algorithms.

2. Objective

The objectives of this project are

- To implement a CNN model using TensorFlow/Keras for classifying handwritten digits.
- To preprocess the MNIST dataset correctly for CNN input (2D spatial shape, normalization, and one-hot encoding).
- To train, validate, and evaluate the model on the standard MNIST train/test split.
- To visualize training performance (accuracy/loss curves) and prediction results.
- To save and reload the trained model for future inference.

3. Dataset

Attribute	Details
Dataset Name	MNIST (Modified NIST)
Source	tensorflow.keras.datasets.mnist
Training Samples	60,000 images
Test Samples	10,000 images
Image Size	28x28 pixels (grayscale)
Classes	10 (digits 0 through 9)
Pixel Range	0 - 255 → Normalized to 0.0 - 1.0
Label Encoding	One-hot vector of length 10

4. Methodology

4.1 Preprocessing

- Images reshaped from $(N, 28, 28) \rightarrow (N, 28, 28, 1)$ to provide the channel dimension required by Conv2D.
- Pixel values normalized from $[0, 255]$ to $[0.0, 1.0]$ by dividing by 255.
- Labels are one-hot encoded using `to_categorical(y, 10)`.

4.2 Model Architecture

A sequential CNN was designed with two convolutional blocks followed by a fully-connected classifier head:

Layer	Config	Output Shape	Parameters
Conv2D	32 filters, 3x3, ReLU	(26, 26, 32)	320
MaxPooling2D	Pool 2x2	(13, 13, 32)	0
Conv2D	64 filters, 3x3, ReLU	(11, 11, 64)	18, 496
MaxPooling2D	Pool 2x2	(5, 5, 64)	0
Flatten	-	(1600)	0
Dense	64 units, ReLU	(64)	102,464
Dropout	rate = 0.2	(64)	0
Dense (Output)	10 units, Softmax	(10)	650

4.3 Training Configuration

Hyperparameter	Value
Optimizer	Adam
Loss Function	Categorical Cross-Entropy
Epochs	10
Batch Size	128
Validation Split	20% of training data
Metrics	Accuracy

5. Results

Metrics	Value
Final Training Accuracy	~99.2%
Final Validation Accuracy	~99.1%
Test Accuracy	~99.0%
Test Loss	~0.03

The CNN model correctly predicted all 10 sample test images with confidence scores of 0.93 - 1.00.

6. Conclusions

This project successfully demonstrated the implementation of a CNN-based image classifier for handwritten digit recognition on the MNIST dataset. Key takeaways include:

- CNNs are far more suitable than fully-connected (Dense) networks for image classification tasks due to their ability to detect spatial features like edges and curves.
- Proper input preprocessing (maintaining 2D shape, normalization, one-hot encoding) is critical for CNN performance.
- The model achieved ~99% test accuracy in just 10 epochs, validating the effectiveness of the chosen architecture.
- Dropout regularization helped prevent overfitting, keeping train and validation accuracy close.
- The trained model was saved in .h5 format and successfully reloaded for inference.

7. AI Usage Declaration

In accordance with the submission guidelines, the following declaration is made regarding the use of AI tools during this project:

AI Tools Used

Tool	Purpose
Claude (Anthropic)	Architecture review, README & report generation

How AI Was Used

- Documentation: Claude generated the README.md and this project report based on the code.

Sections Written / Generated with AI Assistance

- README.md — fully generated by Claude based on project details.
- Project Report (this document) — 50% generated by Claude.